Xunjian Yin

xjyin@pku.edu.cn https://xunjianyin.github.io/

Research Interests

#LLM Evaluation #Self-improvement #Agent #Reasoning

Education

- **Academy for Advanced Interdisciplinary Studies, Peking University** *2022.09-2025.06*
  Master of Computer Science, supervised by Prof. Xiaojun Wan. GPA: 90/100
- **School of Electronics Engineering and Computer Science, Peking University** *2018.09-2022.06*
  Bachelor of Computer Science GPA: 3.7/4.0

Academic Experience

- **NLP Group, University of California, Santa Barbara** *2024.6-2024.09*
  Visiting Student, supervised by Prof. William Wang California
  – Pretrained Reverse LM: LM trained with last token prediction (pretraining and fine-tuning).
  – Gödel Agent: A self-referential agent that can dynamically modify its logic through recursive processes.
- **Microsoft Research Asia** *2022.02-2022.08*
  Research Intern, supervised by Dr. Shuming Ma Beijing
  – Explored the role of Curriculum learning in large language model pre-training.
- **Wangxuan Institute of Computer Technology, Peking University** *2020.10-2022.06*
  Research Assistant, supervised by Prof. Xiaojun Wan Beijing
  – Conducted performance analysis of sequence-to-sequence models for Data2Text tasks.
  – Designed KNN retrieval algorithm to enhance model performance in grammar correction.
- **Institute of Computational Linguistics, Peking University** *2020.04-2021.11*
  Research Assistant, supervised by Prof. Yunfang Wu Beijing
  – Designed a grammar correction model, leveraging dependency parse tree for enhanced performance.

Selected Publications

- **Xunjian Yin**, Xinyi Wang, Liangming Pan, Xiaojun Wan, William Wang. "Gödel Agent: A Self-Referential Framework for Agents Recursively Self-Improvement". *submitted to ICLR 2025*
- Xu Zhang*, **Xunjian Yin***, Xiaojun Wan. "ContraSolver: Self-Alignment of Language Models by Resolving Internal Preference Contradictions". *submitted to NAACL 2025*
- **Xunjian Yin***, Xu Zhang*, Jie Ruan, Xiaojun Wan. "Benchmarking Knowledge Boundary for Large Language Model: A Different Perspective on Model Evaluation". *ACL 2024*
- **Xunjian Yin**, Jin Jiang, Liming Yang, Xiaojun Wan. "History Matters: Temporal Knowledge Editing in Large Language Model". *AAAI 2024*
- **Xunjian Yin***, Baizhou Huang*, Xiaojun Wan. "ALCUNA: Large Language Models Meet New Knowledge". *EMNLP 2023*
- **Xunjian Yin**, Xiaojun Wan. "How Do Seq2Seq Models Perform on End-to-End Data-to-Text Generation?" *ACL 2022*

Awards

- **Award for Academic/Research Excellence, Merit Student, PKU** *2019, 2020, 2022, 2023, 2024*
- **Wang Xuan Scholarship, Huatai Securities Scholarship, PKU** *2022, 2024*

Services

- **Teaching Assistant:**
  – Introduction to Computing 2021, Data Structures and Algorithms 2022, Web Data Mining 2023
- **Reviewer:**
  – ACL, EMNLP, NAACL, NeurIPS, ICLR, ACL Rolling Review
- **Volunteer:**
  – AAAI 2024, ACL 2024